



Broadband and compact polarization beam splitter in LNOI hetero-anisotropic metamaterials

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Abstract: In this paper, theoretical modeling and numerical simulations of a high-performance polarization beam splitter (PBS) based on hetero-anisotropic metamaterials are proposed on the lithium-niobate-on-insulator (LNOI) platform. The hetero-anisotropic metamaterials constructed by sub-wavelength gratings (SWGs) can be regarded as effective anisotropy medium, which exhibits strong birefringence without breaking the geometrical symmetry, contributing to the formation of PBS. Rather than the principle of PBS based on beat-length difference of transverse electric (TE) polarization and transverse magnetic (TM) polarization, the device can realize polarization beam splitting in single beat length, and the footprint of the proposed PBS can be reduced to $8\ \mu\text{m} \times 160\ \mu\text{m}$ (with S-bend). The simulation results show that the bandwidth is 185 nm (1450~1634 nm) for TE polarization while the bandwidth is 85 nm (1490~1575 nm) for TM polarization when the polarization extinction ratio is $>20\ \text{dB}$. Furthermore, the insertion loss is less than 1 dB in the range of 1450 to 1650 nm, for both TE and TM polarization. Additionally, the proposed device proves strong robustness of the fabrication tolerance.

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1. Introduction

Lithium-niobate (LN) offers excellent pyroelectric, nonlinear, ferroelectric and photorefractive properties [1,2]. It is worth mentioning that due to the high electro-optical coefficient of LN, photonic integrated circuits (PICs) based on LN have been widely exploited to develop numerous applications, such as modulators [3,4], optical switches [5] and other tunable devices [6,7]. However, nearly all the LN nano-photonic devices are polarization dependent owing to its birefringence and tensor electro-optic coefficient. A polarization diversity scheme has been proposed as an effective approach to overcome the limits in the PICs. In the scheme, polarization beam splitter (PBS) is one of the key components to separate the polarization state of light.

Over the last decades, the fabrication of LN-based PBS has relied on titanium diffusion, resulting in a low refractive index difference between the core and cladding layers (about 0.02), so that the footprint of PBS is generally in several millimeters or more than 10 millimeters [8–12]. Recently, the improvement of crystal ion slicing for thin film and dry etching technology has made it possible to fabricate compact PBS on lithium-niobate-on-insulator (LNOI) platform [13]. For instance, the PBS can be realized in photonic crystal structure which has different transmission conditions for transverse electric (TE) polarization and transverse magnetic (TM) polarization [14]. This PBS can achieve broad angle (8°) and bandwidth (BW~70 nm) with the polarization extinction ratios (PER) higher than 10 dB. Instead, PBS with high PER ($>20\ \text{dB}$), large BW (~200 nm) and low insertion loss (IL<0.9 dB) can be realized based on the principle of asymmetric Mach-Zehnder interferometer (MZI) [15]. In this MZI-based PBS, the phase

shift of TE polarization and TM polarization can be controlled simultaneously by combining multi-mode interference (MMI) couplers and delay lines, so that the incident TE and TM modes can be coupled to different output ports. However, due to the long delay lines and MMI couplers in asymmetric MZI structure, its length has not been limited in sub-millimeter. In order to further reduce the device footprint, PBS based on hybrid platform is proposed. Benefitting from the significant phase mismatch caused by assisted silicon nitride [16] or silicon [17], the PBS based on LNOI can be realized in sub-millimeter now. Nevertheless, it is still a great challenge to fabricate waveguide with heterogeneous structures. Ideally, a PBS should feature a low IL, large BW, compact footprint and simplification of manufacturing procedure.

Synthesis of metamaterials provide a novel approach to realize PBS with controllable index profiles and dispersion properties [18,19]. Among them, sub-wavelength grating (SWG) is an imperative and efficient structure to enable the synthesis of metamaterial because of its strong anisotropy and flattened dispersion [20–23]. High performance silicon PBSs with SWG structure have been intensively proposed based on MMI [24,25], asymmetrical directional coupler (ADC) [26,27], MZI [18] in the last years. In this paper, we propose and analyze a compact PBS based on LNOI hetero-anisotropic metamaterials in sub-millimeter. By assembling SWGs with different orientations, significant birefringence is achieved through the effective medium anisotropy rather than the configuration asymmetry mentioned in the previous articles. Inspired by this effect, the device can show different transmission functions (e.g. straight waveguide, MMI, ADC) for different polarization. The MMI-based PBS can provide broad BW and high PER, but suffer from the issue of large device footprint [19]. The ADC-based PBS can achieve small footprint, but BW is limited. Here, with the help of flattened dispersion of SWG and asymmetrical structure, the BW of the proposed ADC-based PBS can be broadened. The ADC-based PBS consists of a pair of hetero-anisotropic strip waveguides, which can be implemented to separate TE polarization and TM polarization. For TE polarization, the hetero-anisotropic strip waveguides behave as two isolated waveguides. For TM polarization, the hetero-anisotropic strip waveguides behave as an adiabatic coupler. The simulation results show that for TE polarization, the IL is about 0.3 dB, and the PER is higher than 20 dB in 185 nm BW. For TM polarization, the IL is about 1.0 dB, and the PER is higher than 20 dB with the BW of 85 nm. Therefore, a PBS with low IL, high PER, broadband width and small footprint can be achieved.

2. Structure and design principle

Figure 1(a) shows the configuration of proposed LNOI-based PBS, which consists of hetero-anisotropic metamaterials. The device contains two oriented SWGs. The SWGs with optical axis parallel to x -/ y -direction are represented as SWG $\perp$ / SWG $\parallel$, as shown in Fig. 1(a). According to the effective medium theory, SWG can be calculated as an equivalent uniform medium [20,25,28,29], and its effective refractive index can be expressed as follows:

$$n_o^2 = f \cdot n_1^2 + (1 - f) \cdot n_2^2 \quad (1)$$

$$\frac{1}{n_e^2} = \frac{f}{n_1^2} + \frac{1-f}{n_2^2} \quad (2)$$

where n_1 and n_2 are the refractive indices of lithium niobate and silicon dioxide, f is the SWG duty cycle. n_o and n_e are the ordinary and extraordinary refractive indices. The refractive index tensors for SWG $\perp$ and SWG $\parallel$ can be written as follows [20,25,28,29]:

$$n_{\text{SWG}\perp} = \text{diag}[n_e, n_o, n_o] \quad (3)$$

$$n_{\text{SWG}\parallel} = \text{diag}[n_o, n_e, n_o]. \quad (4)$$

As shown in Fig. 1(b), hetero-anisotropic metamaterials have different refractive indices for TE polarization and TM polarization. In detail, for TE polarization, the refractive indices of SWG $\perp$

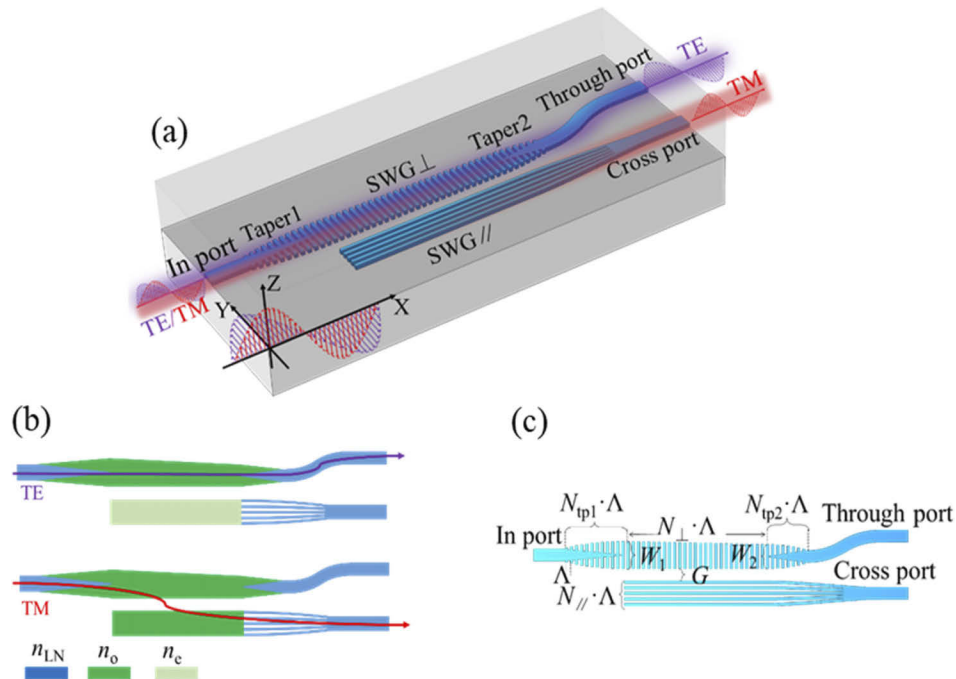


Fig. 1. Schematic of the proposed PBS on LNOI waveguide. (a) 3D view of the proposed PBS. (b) Working principle for hetero-anisotropic metamaterials PBS. (c) The top view of the proposed PBS.

and SWG// are n_o and n_e , respectively, which breaks the phase matching condition required for the coupling between SWGs. Therefore, when TE mode is injected, the device will display as two isolated waveguides in a large BW. For TM polarization, the refractive indices of SWG⊥ and SWG// are equivalent to n_o , and the device behaves as an adiabatic coupler.

To reduce fabrication difficulties, the manufacturing process should be facilitated and the minimum feature size of the device should be maximized. In this paper, single-etch is essential in the fabrication and the duty cycle is set to be 50%. As shown in Fig. 1(c), to make a trade-off between good confinement of optical mode and the minimum fabrication accuracy of dry etching [20,30], the period of SWG⊥/SWG// is 360 nm. In order to satisfy the single-mode transmission condition of waveguide, the width of both input and output waveguides are 1 μm . For reducing the back reflection and the loss caused by mode mismatch between the straight waveguide and the SWG⊥, a tapering structure, which consists of a tapered waveguide and a tapered SWG, are proposed. In addition, multi-segment S-shaped waveguides are introduced to reduce the loss due to mode mismatch between the SWG// and the cross port. For the sake of avoiding the coupling between the through port and the cross port, a S-shaped waveguide is fixed up in front of the through port.

3. Simulated results and discussion

In this paper, we simulate the proposed PBS based on 250 nm LN film by using 3D finite-difference-time-domain method (FDTD, Ansys-Lumerical FDTD Solution). Before analyzing the performance of PBS devices, there are several structural parameters that directly affect the loss of PBS need to be discussed first. Firstly, we analyze the loss caused by the taper between the input waveguide and SWG⊥. As shown in Fig. 2(a), in this case the width of input waveguide (W_{in}) is 1 μm , and the width of SWG⊥ (W_1) is 1.55 μm .

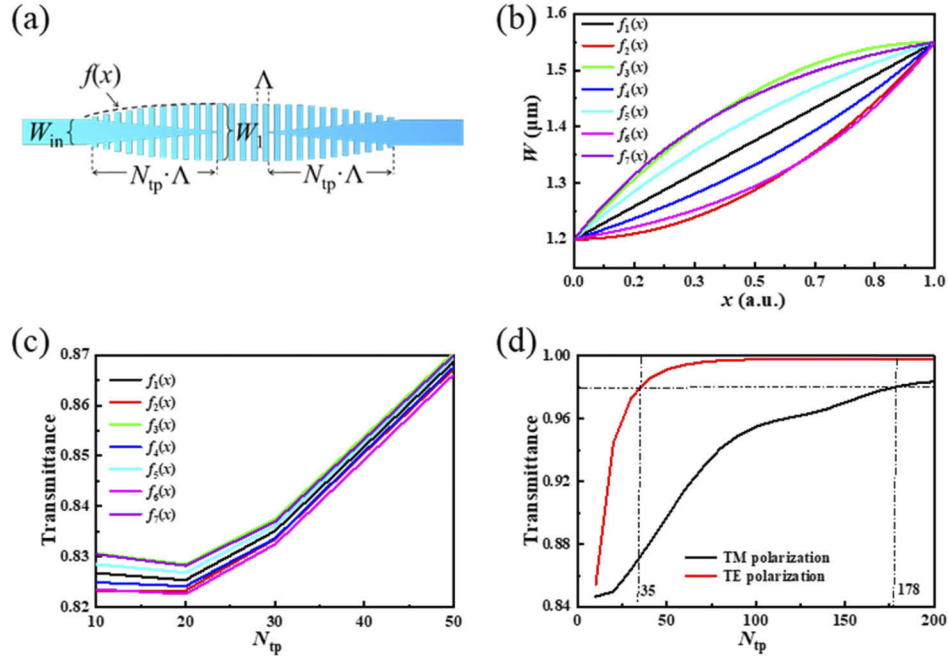


Fig. 2. (a) Schematic illustration of SWG-based taper. (b) Envelope function of SWG width. (c) The transmission spectra of TM polarization corresponding to different period number N_{tp} . (d) The transmission spectra as a function of period number N_{tp} when tapered-SWG width distribution is $f_3(x)$.

As shown in Fig. 2(a), we calculated the IL of the taper with envelope function $f(x)$ and period number N_{tp} . The width variation curves of tapers are shown in Fig. 2(b), and the expressions are as follows.

$$\begin{aligned}
 f_1(x) &= W_{in} + x(W_1 - W_{in}) \\
 f_2(x) &= W_{in} + x^2(W_1 - W_{in}) \\
 f_3(x) &= W_1 - (1 - x)^2(W_1 - W_{in}) \\
 f_4(x) &= W_{in} + (e^x - 1)(W_1 - W_{in})/(e - 1) \\
 f_5(x) &= W_1 - (e^{1-x} - 1)(W_1 - W_{in})/(e - 1) \\
 f_6(x) &= W_{in} + (e^{2x} - 1)(W_1 - W_{in})/(e^2 - 1) \\
 f_7(x) &= W_1 - (e^{2-2x} - 1)(W_1 - W_{in})/(e^2 - 1)
 \end{aligned} \tag{5}$$

Here x is the relative length of $N_{tp} \cdot \Lambda$. The IL in TM mode of the structure in Fig. 2(a) is shown in Fig. 2(c). It can be seen from Fig. 2(c) that the taper corresponding to the envelope function $f_3(x)$ has the minimum IL. Therefore, the width distribution of tapers between straight waveguide and SWG $\perp$ in this paper is designed as $f_3(x)$. As shown in Fig. 2(d), we analyze the transmission in TE/TM mode of SWG taper with different N_{tp} . One sees that a small N_{tp} (<30) cannot evolve effectively the mode distribution and thus gives a high loss. On the other hand, a large N_{tp} will increase the device size. Thus, a trade-off should be made when choosing an appropriate N_{tp} . To achieve near lossless transmission ($>98\%$), the minimum N_{tp} for TE polarization is 35, while those for TM polarization is 178. It should be noted that both TE and TM polarization will pass through Taper1 (in Fig. 1(a)). However, only TE polarization will pass through Taper2 (in

Fig. 1(a)). Therefore, considering the tradeoff between the total length and the IL of the PBS, we set N_{tp1} and N_{tp2} to be 178 and 35, respectively.

Then, the S-bend between SWG// and cross port in Fig. 1 will also cause additional loss. Here a model is established to analyze it, as shown in Fig. 3(a). As only TM mode is coupled to SWG//, we analyze the loss characteristics of TM in S-bend waveguides with different lengths (L_{S-bend}). As shown in Fig. 3(b), with the increasing L_{S-bend} , the transmission efficiency will be higher. To balance the length and IL of PBS, L_{S-bend} is selected as 71.3 μm .

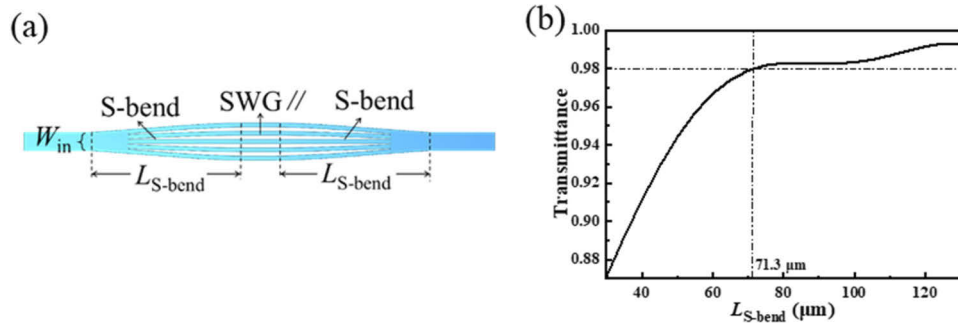


Fig. 3. (a) Schematic illustration of S-bend connector. (b) The transmission spectra of TM polarization corresponding to different L_{S-bend} .

Also, the structure parameters are optimized by particle swarm optimization. The optimized parameters are as follows: $G = 0.8 \mu\text{m}$, $W_1 = 1.7 \mu\text{m}$, $W_2 = 0.2 \mu\text{m}$, the cycles of SWG $\perp$ and SWG// are 189 and 5 respectively. The optical transmission and spectrum of TE and TM polarization corresponding to the device can be obtained, as shown in Fig. 4. It can be seen from Fig. 4(a) and (b) that the incident TE polarization is directly guided to the through port. However, the incident TM polarization is coupled to SWG// and output from the cross port. More detailed information can be obtained from Fig. 4(c) and (d). For TE mode, the IL is less than 0.3 dB and

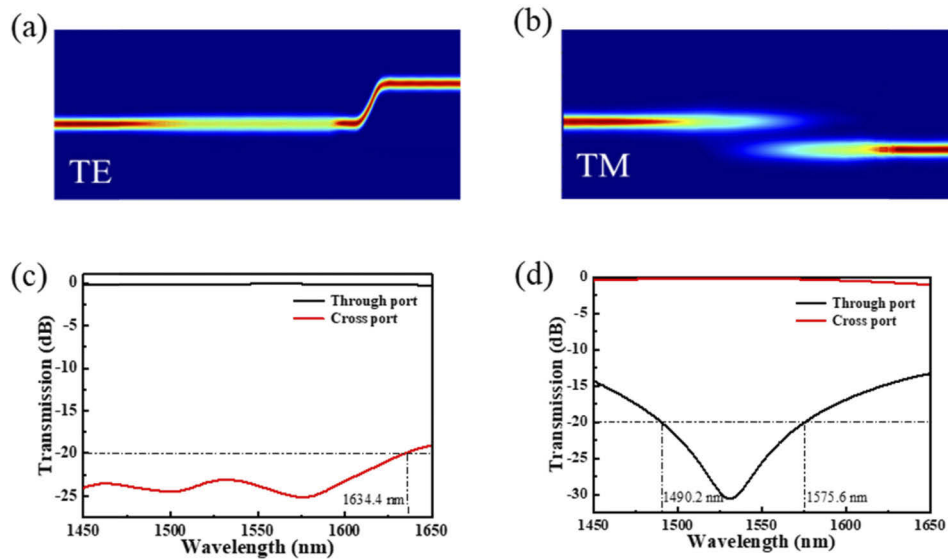


Fig. 4. Simulated distributions of the light in (a) TE mode and (b) TM mode at wavelength of 1550 nm. The calculated transmittance spectra in (c) TE mode and (d) TM mode.

the PER is greater than 20 dB at 1450.0~1634.4 nm; for TM mode, the IL is less than 1.0 dB and the PER is greater than 20 dB at 1490.2~1575.6 nm.

In addition, the fabrication tolerance is calculated for the proposed device. The schematic diagram is shown in Fig. 5. The fabrication errors include the deviation of waveguide width δw , the SWG nano-strip δl , the LN thin film thickness δh , and waveguide sidewall angle $\delta\theta$ (see

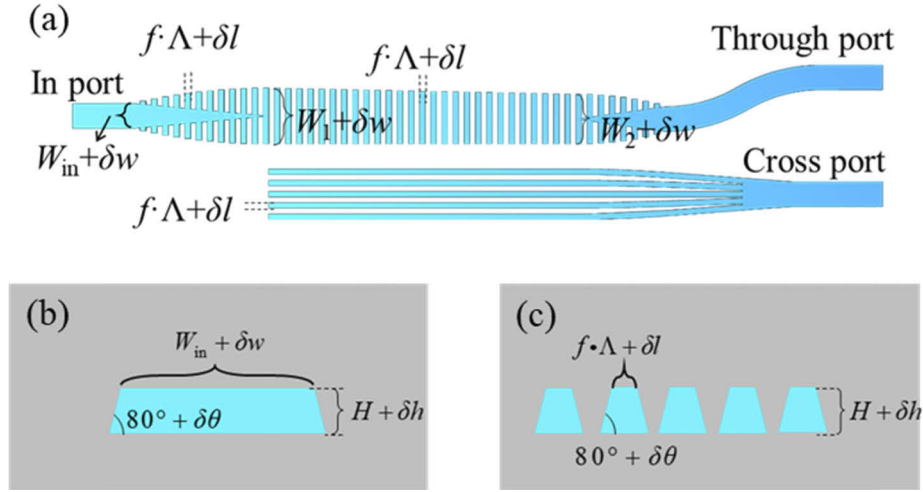


Fig. 5. (a) Top view of the PBS with fabrication tolerance. Side view of the straight waveguide (b) and SWG (c) with fabrication tolerance.

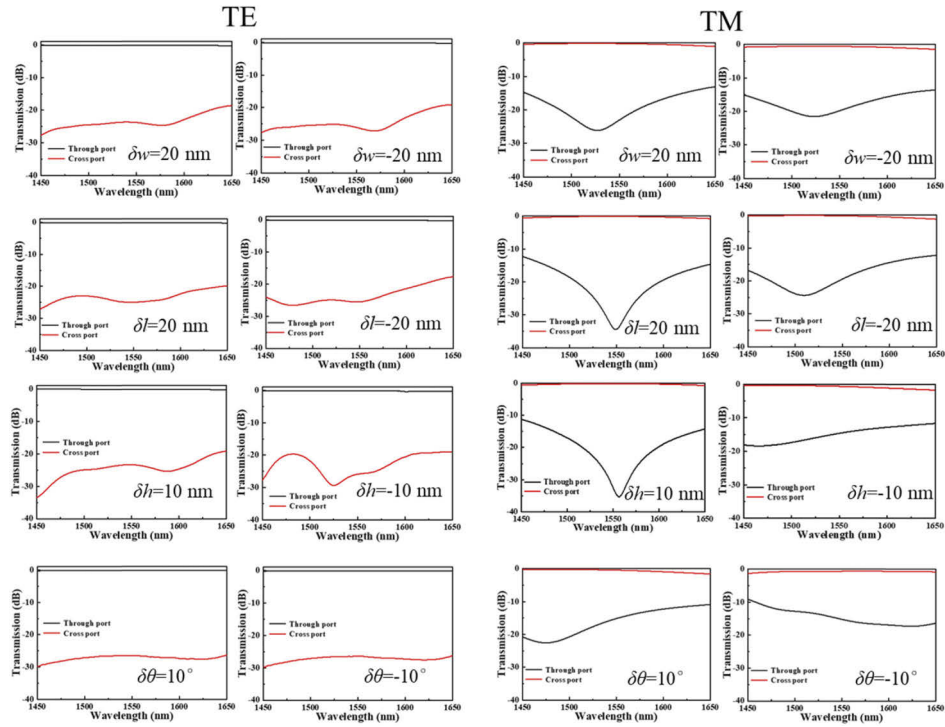


Fig. 6. The calculated transmittance spectra for the PBS with deviated parameters.

Fig. 5). It can be seen from Fig. 6 that the PER > 15 dB and IL < 1 dB can be obtained over BW > 150 nm when $\delta w/\delta l$ are varied in the range of ± 20 nm, or when δh is varied in the range of ± 10 nm. It should be noted that it is still difficult to fabricate LN waveguides with completely vertical sidewalls [31,32]. The simulation results show that the PER > 10 dB and IL < 1 dB can be obtained over BW > 200 nm when $\delta\theta$ fluctuates in the range of $\pm 10^\circ$. The strong robustness to fabrication errors of the proposed device is benefit to reduce the fabrication difficulties.

4. Conclusion

In this paper, a novel LNOI PBS based on hetero-anisotropic metamaterials is proposed and simulated. The hetero-anisotropic metamaterials, constructed by SWGs in different directions, presented as different refractive indices for TE/TM polarization. Due to the anisotropic properties of different polarization, the length of PBS based on SWGs can be limited to one single beat length, rather than the least common multiple of TE and TM beat length. Meanwhile, benefitting from the low dispersion of SWG, the device exhibits a large operating BW which can cover almost S + C+L band. Furthermore, the IL of the device is less than 1.0 dB, and the PER is higher than 20 dB. In addition, the simulation results show that the device performances are not significantly affected by the fabrication tolerance. We believe such a device could play a significant and valuable role in coherent optical communication, polarization diversity system.

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